Example 5.9 Find the value of 'a' such that $PQ = QR$ where P, Q, and R are the points whose coordinates are (6, –1), (1, 3) and (a, 8) respectively.

Solution

Given P (6, –1), Q (1, 3) and R (a, 8)

$$PQ = \sqrt{(1-6)^2+(3+1)^2} = \sqrt{(-5)^2+(4)^2} = \sqrt{41}$$

$$QR = \sqrt{(a-1)^2+(8-3)^2} = \sqrt{(a-1)^2+(5)^2}$$

$$\text{Given } PQ = QR$$

$$\text{Therefore } \sqrt{41} = \sqrt{(a-1)^2+(5)^2}$$

$$41 = (a-1)^2 + 25 \quad \text{[Squaring both sides]}$$

$$(a-1)^2 + 25 = 41$$

$$(a-1)^2 = 41 - 25$$

$$(a-1)^2 = 16$$

$$(a-1) = \pm 4 \quad \text{[taking square root on both sides]}$$

$$a = 1 \pm 4$$

$$a = 1 + 4 \text{ or } a = 1 - 4$$

$$a = 5 \text{ or } a = -3$$

Example 5.10 Let $A(2, 2)$, $B(8, -4)$ be two given points in a plane. If a point P lies on the X- axis (in positive side), and divides AB in the ratio 1: 2, then find the coordinates of P.

Solution

Given points are $A(2, 2)$ and $B(8, -4)$ and let $P = (x, 0)$ [P lies on x axis]

By the distance formula

$$d = \sqrt{(x_2-x_1)^2+(y_2-y_1)^2}$$

$$AP = \sqrt{(x-2)^2+(0-2)^2} = \sqrt{x^2-4x+4+4} = \sqrt{x^2-4x+8}$$

$$BP = \sqrt{(x-8)^2+(0+4)^2} = \sqrt{x^2-16x+64+16} = \sqrt{x^2-16x+80}$$

Given $AP : PB = 1 : 2$

i.e. $\frac{AP}{BP} = \frac{1}{2} \quad (\because BP = PB)$

$$2AP = BP$$

squaring on both sides,

$$4AP^2 = BP^2$$

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